

Jorge Emanuel Nuñez

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Education

- San Diego State University | San Diego, CA

May 2027

 - M.S. in Computer Science, GPA: 3.75
 - Thesis:** Researching the use of 3D scene graphs to improve 3D multi-object tracking and long-term object identity association for autonomous vehicles.
 - Coursework: Machine Learning, Reinforcement learning, Data Mining, Computer Networks & Distributed Systems
- University of California, Berkeley | Berkeley, CA

May 2024

 - B.S. in Electrical Engineering and Computer Sciences
 - Coursework: Operating Systems, Artificial Intelligence, Database Systems, Algorithms, Computer Architecture

Experience

- Computer Systems Intelligence Lab, SDSU | Graduate Research Assistant

October 2025 - Present

 - Deployed NVIDIA Isaac ROS Visual SLAM on a Jetson Orin Nano (JetPack 6/L4T R36) in a Docker-based ROS 2 Humble environment, fusing stereo camera + IMU data to replace AMCL as the localization source for Nav2 autonomous navigation.
 - Built a Nav2-compatible 2D occupancy map from 3D vSLAM landmarks (Z-axis filtering, Bresenham ray casting, morphological operations), validated on the EuRoC MH_01 dataset via rosbag replay, achieving autonomous goal navigation in RViz2.
- Microsoft | Software Engineering Intern

May 2026 - August 2026

 - Built and shipped ~10 UI components in TypeScript/React for M365 Copilot's enterprise customer base, integrating with backend services and telemetry pipelines.
 - Instrumented telemetry logging to track feature adoption and surface usage insights across the Copilot experience.
 - Partnered with a 40-person engineering team and 10–20 partner teams (PM, EM, design) to align on requirements, compliance, and release processes.
- Qualcomm | Software Engineering Intern

May 2023 - August 2023

 - Contributed to the CPU software infrastructure team responsible for validating and optimizing low-level system performance in embedded platforms.
 - Developed and maintained Python-based automation tools to validate power management workflows, analyzing large datasets of system performance metrics to identify regressions.
 - Automated CI/CD testing with Bash and CMake, reducing deployment time by 40%.
 - Worked in a 5-person Agile team to deliver weekly builds with optimized power-saving workflows.
 - Reduced system power consumption by analyzing CPU load patterns and optimizing task scheduling.
- Berkeley ACE Lab | Research and Development Assistant

February 2021 - May 2024

 - Engineered and implemented RESTful APIs using Node.js/TypeScript to integrate real-time cheat detection with PrairieLearn, handling 200+ concurrent users.
 - Developed Python-based cheat-detection software with 92% accuracy, reducing academic dishonesty in UC Berkeley's introductory computer science course serving 200+ students.
 - Launched automated test suites and CI/CD pipelines for continuous deployment, ensuring 99% system uptime.
 - Authored documentation and optimized Git workflows, boosting collaboration for a 12-member team.

Projects

- Traffic Sign Recognition | Personal Project

 - Achieved 97%+ accuracy on GTSRB with a LeNet5-inspired CNN in PyTorch, trained on augmented data using cross-entropy loss and Adam optimizer.
 - Applied image preprocessing and data augmentation (histogram equalization, normalization) to improve model robustness to real-world variation
- Pintos Operating System | Course Project, CS162: Operating Systems

 - Extended OS with user programs, system calls, priority scheduling, and file caching in C.
 - Implemented robust synchronization primitives, including semaphores and priority donation to prevent priority inversion in multi-threaded processes.
- Database Management System | Course Project, CS186: Database Systems

 - Built an ACID-compliant DBMS in Java with MVCC and ARIES-style crash recovery.
 - Reduced lookup time by 60% using custom B+ Tree indexing and optimized query plans.

Skills

- ML & Computer Vision:** Pytorch, CNNs, CUDA, NumPy, Pandas
- Robotics and Perception Systems:** ROS2, Nav2, NVIDIA Isaac ROS, Sensor Fusion, V-SLAM
- Languages:** Python, C/C++, Java, TypeScript/JavaScript, SQL, React
- Systems & Infra:** Docker, Git, CI/CD pipelines, Linux, Restful APIs